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Design and Development of a Vision-based Robotic Arm Banana Sorting System for Quality Control of Banana Ketchup Manufacturing

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Abstract- Automation of quality control processes in food manufacturing are essential for enhancing efficiency, consistency, and productivity. This study presents the design and development of a vision-based robotic arm banana sorting system for quality control in banana ketchup manufacturing. The system integrates computer vision, artificial intelligence, and robotics to classify bananas according to industry standards. Using the YOLOv11 deep learning model, the system categorizes bananas into good (Extra Class, Class 1, and Class 2) and bad (bananas that do not meet quality standards). The classification process relies on image processing techniques implemented in PyCharm, using cv2, cvzone, threading, serial, time, and servo libraries. A robotic arm, controlled by servo motors and an Arduino microcontroller, performs pick-and-place operations based on the AI model's classification results. The system computes 3D coordinates from 2D camera data, allowing object localization and movement. Real-time communication among the camera, laptop, and Arduino microcontroller is maintained through a USB 3.0 connection to support data transfer and robotic control. Evaluation of different YOLOv11 variants (YOLOv11n, YOLOv11m, and YOLOv11x) indicated that YOLOv11m provided a suitable balance of accuracy and speed for real-time sorting. Experimental results demonstrated classification accuracy and sorting performance, confirming the system's reliability in automating banana quality assessment. The findings indicate the potential of AI-driven robotic sorting to enhance food processing efficiency, reduce manual labor, and maintain consistent product quality. This research contributes to the development of automation in the agricultural and food manufacturing industries and supports further work in AI-based quality control systems.

Keywords—Robotics, Artificial Intelligence, Robotic Arm, Banana Sorting, YOLOv11, Quality Control, Food Automation

In recent years, the demand for banana ketchup has surged, fueled by shifting consumer preferences toward unique and healthier condiment options. Industry projections indicate that the global banana ketchup market is anticipated to reach USD 1.23 billion by 2023, with a compound annual growth rate (CAGR) of 5.4% from 2024 to 2030. [1] This increasing demand presents significant challenges for manufacturers, particularly in scaling their operations without a proportional increase in labor costs.

Current practices in banana ketchup production heavily rely on manual sorting of bananas, which is labor-intensive and time-consuming. [2] This inefficiency becomes particularly pronounced during peak production periods, leading to substantial delays in processing. Moreover, the lack of standardized measures for sorting bananas often results in inconsistencies across batches, ultimately impacting product quality and diminishing consumer trust. [3] Additionally, the COVID-19 pandemic has highlighted the vulnerabilities of manual sorting systems, as improper handling and sorting can elevate risks related to food safety, contamination, and spoilage of raw materials. [4]

Automation and robotics present viable solutions to these challenges. Research indicates that traditional manual sorting is not only labor-intensive but also prone to errors stemming from human fatigue and variability in inspector experience. [5] AI-based systems can automate critical sorting tasks such as evaluating maturity, identifying quality, and detecting defects in bananas. Furthermore, advancements in computer vision and machine learning have enabled precise quality assessments based on key parameters, including size, ripeness, and defects. [6] These innovative technologies are increasingly adopted in food manufacturing to streamline quality control processes, enhance productivity, and reduce waste.

I. INTRODUCTION

This study aims to design and develop an automated robotic banana sorting system specifically for banana ketchup production. By integrating robotics, machine learning algorithms, and computer vision technologies, the proposed system will objectively evaluate key quality parameters essential for production. The overarching goal is to enhance efficiency, minimize waste, and ensure that only high-quality bananas are selected for processing. This initiative aligns with findings by Samuel Barasa et al. (2023), who emphasized the critical role of robotics in maintaining consistent quality standards and improving productivity in food manufacturing. [8]

Through this research, the integration of robotics and artificial intelligence in food processing is positioned as a progressive solution to the limitations of traditional quality management methods, paving the way for innovation and sustainability within the industry.

II. METHODOLOGY

This chapter presents the design, development, tools, and methodologies employed in constructing a prototype for the vision-based robotic arm banana sorting system aimed at enhancing quality control in banana ketchup manufacturing.

A. Project Design



Figure 1. System Architecture

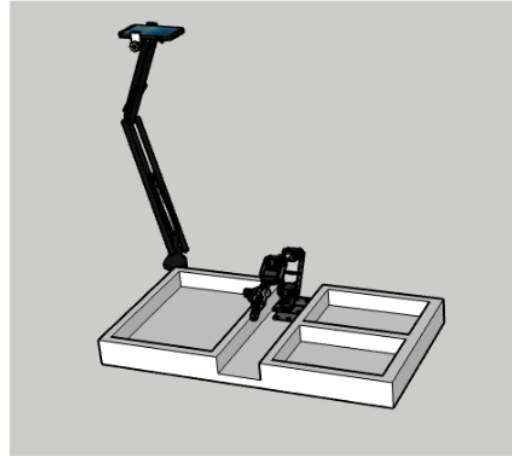
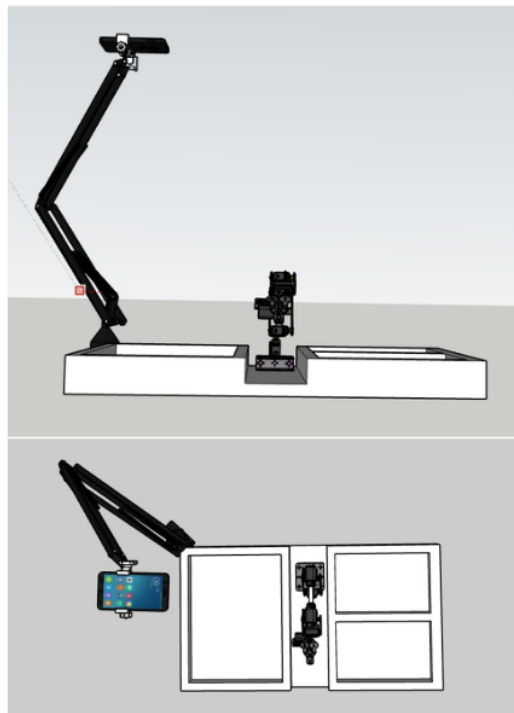


Figure 2. 3D Design of Vision-based Robotic Arm

The proposed system consists of four major components: a camera, a microprocessor, a microcontroller, and a robotic arm, as illustrated in Figure 1. The first step in the process involves the camera capturing an image of the banana, which serves as the input for the system. Once the image is captured, it is transferred to the AI model for analysis. The AI algorithm processes the image to classify the banana based on its ripeness and to calculate its 3D coordinates. Following this, the processed data, including the classification and 3D coordinates, is transmitted to the microcontroller via serial communication. The microcontroller then interprets the data and sends control signals to the robotic arm's servo motors. These signals are communicated through Pulse Width Modulation (PWM), allowing the microcontroller to precisely control the movement of the robotic arm's end-effector. This enables the robotic arm to move and position itself accurately to grasp the banana at the calculated coordinates, facilitating the subsequent handling or processing steps in the system.

B. Project Development

1. Hardware Development

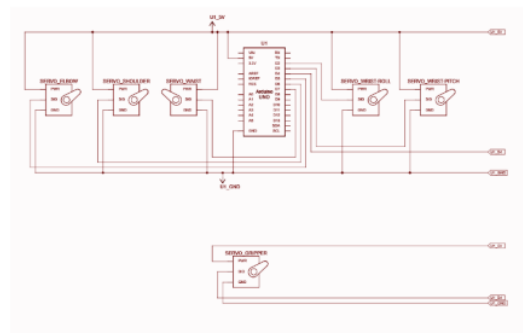


Figure 3. Arduino Circuit connection for Servo Motor



Figure 4. Connection of Smartphone, Laptop, and Arduino

The robotic arm is powered by six MG996R servo motors, each featuring a three-pin connector (orange: signal, brown: ground, red: 5V DC power). These motors control the waist, shoulder, elbow, wrist roll, wrist pitch, and gripper joints. Each servo motor is connected to an Arduino microcontroller, which receives pulse-width modulation (PWM) signals to adjust the motor positions. The synchronized operation of these motors ensures precise and dynamic control of the robotic arm, enabling it to perform complex manipulation and positioning tasks. The careful wiring and integration with the Arduino system guarantee reliable and efficient servo motor functionality.

The smartphone acts as the vision sensor to the laptop. Image data from the smartphone is transmitted to the laptop, where an artificial intelligence (AI) algorithm classifies the objects and calculates the three-dimensional (3D) coordinates of the detected bananas. The laptop communicates the calculated joint angles for positioning the robotic arm to the Arduino via a USB 3.0 Type-A to Type-B cable. Using this data, the Arduino generates the appropriate PWM signals for each servo motor, enabling the robotic arm to move with precision and accuracy to pick and place bananas according to their classification.

The system operates seamlessly over a shared Wi-Fi network between the smartphone and laptop, while the USB 3.0 connection between the GPU and microcontroller ensures fast data transfer and efficient control of the robotic arm. This integrated setup enables autonomous and efficient sorting of bananas.

2. Software Development

PyCharm

A powerful IDE used to develop and debug the Python script that implements the Denavit-Hartenberg (D-H) convention for forward kinematics calculations. This script is crucial for determining the joint angles required to position the robotic arm's end-effector at specific locations.

Arduino IDE

A user-friendly development environment for programming microcontrollers, specifically used here to control the servo motors that actuate the 6-DOF robotic arm. By sending precise commands to the servos, the Arduino ensures accurate and coordinated movements of the arm.

YOLOv11

A state-of-the-art object detection algorithm employed to identify and classify "Saba" banana objects within images captured by the robot's vision system. This algorithm's high accuracy and speed are essential for real-time object detection and localization.

Serial Communication

A method of transmitting data between electronic devices, in this case, between the Arduino microcontroller and the laptop. This communication channel enables the exchange of control signals and sensory data, ensuring seamless integration between hardware and software components.

Label Studio

Label Studio is an open-source data labeling tool that supports diverse data types and annotations, integrating seamlessly with machine learning models for predictive labeling and active learning. It streamlines the development, training, and deployment of computer vision models. In this study, it was used to annotate 300 images into "good" and "bad" classes, with the data exported as a .yaml file for YOLOv11 model training.

C. Testing Procedure

Integration and Operation

The vision system captures images of bananas on the platform, which are then analyzed by image processing software to identify defects and classify the bananas. Based on the analysis results, the control system generates commands for the robotic arm, directing it to pick up the classified banana and place it in the designated sorting bin. The system operates continuously in real-time, ensuring efficient and accurate sorting by constantly processing and sorting bananas.

AI Vision Testing

AI vision testing involves capturing images of objects or subjects from various angles to evaluate the system's ability to process and analyze visual data in diverse perspectives. High-resolution photographs are taken from different viewpoints, including frontal, side, and top-down angles, ensuring that the AI model can accurately detect and classify key features, dimensions, and movements regardless of orientation.

Gripper Testing

Gripping testing for bananas in a sorting system involves evaluating the robotic or mechanical system's ability to accurately and gently grip and classify bananas into two categories: "good" and "bad." The system is designed to apply the right amount of force to securely hold bananas without damaging them, while assessing their quality based on set criteria, such as ripeness, blemishes, and shape. Bananas are then directed to their designated bins: one for good quality bananas and the other for bad quality ones. The test ensures that the system can reliably and consistently identify and separate the bananas based on their classification, ensuring optimal handling and sorting efficiency.

Quality Control and Optimization

Component testing involves rigorous evaluation of individual parts to ensure their optimal performance and reliability. System integration testing follows, comprehensively testing the entire system to verify seamless communication and coordination between both hardware and software components. Regular calibration of the vision system and robotic arm is crucial to maintain accuracy and precision in operations. Additionally, continuous monitoring and troubleshooting through debugging and optimization

processes help identify and resolve issues, ultimately enhancing overall system performance.

D. Performance Assessment

a. Method of Evaluation

To assess the detection capabilities of the Yolov11 model, we employed several classification metrics: accuracy, recall, precision, F1-score, and mean average precision (mAP). Label Studio provided accuracy, recall, and precision metrics following dataset training. To evaluate the model further, the researchers will manually compute the F1-score and mean average precision (mAP) using the formulas provided below.

Equation 1: F-1 Score Formula

$$F1 \text{ Score: } F1 = 2 \times \frac{PRECISION \times RECALL}{PRECISION + RECALL}$$

Equation 2: Mean Average Precision Formula

$$mAP = \frac{1}{n} \sum_{k=1}^n AP$$

b. Confusion Matrix

To evaluate the robotic arm's banana classification accuracy, the study employed a confusion matrix. This matrix effectively summarizes the model's performance by quantifying the number of four possible outcomes:

- **True Positives (TP):** Bananas were correctly classified as bananas.
- **True Negatives (TN):** Non-banana objects were correctly classified as non-bananas.
- **False Positives (FP):** Non-banana objects were incorrectly classified as bananas (false alarms).
- **False Negatives (FN):** Bananas were incorrectly classified as non-bananas (missed detections).

This analysis was crucial in determining the system's ability to accurately categorize bananas based on their quality (e.g., ripe, unripe, defective). The insights derived from the confusion matrix provided a comprehensive understanding of both correct and incorrect classifications, facilitating the identification of areas for improvement within the sorting algorithm.

c. Testing Validation

To validate the sorting system, a series of tests were conducted using randomly selected batches of bananas, categorized by quality attributes. The results of these experiments were analyzed using established performance equations, enabling a comprehensive evaluation of the robotic arm's performance compared to manual sorting methods. This systematic approach ensured a thorough examination of all aspects of the sorting process, leading to reliable conclusions regarding the system's effectiveness.

Equation 3. Percentage of Correct Classifications, %CC

$$\%CC = \frac{\text{Number of all Correct Classifications}}{\text{Total Number of Samples}} \times 100\%$$

Table 1. AI Classification Testing

Trial	Input Description	Pickup Result	Drop Result	Accuracy
1	Extra Class, Class 1, Class 2			

Percentage of Correct Classification	%
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The performance of the robotic banana sorter, particularly its pick-and-place functionality, was evaluated as shown in Table 2. The methodology measured the system's accuracy in grasping and depositing bananas into the "Good" and "Bad" trays. A series of test runs were conducted, with bananas placed at various slots within the system's operational range. Equation 4 calculates the success rate of pick-and-place operations as a percentage, reflecting the system's accuracy and efficiency in sorting bananas.

Equation 4. Percentage of Successful Pick & Drop, %PD

$$\%PD = \frac{\text{Number of all Successes}}{\text{Total Number of Slots}} \times 100\%$$

Table 2. Pick and Drop Points Testing

Trial	Slot	Pick		Drop	
		Pick Points	Results	Drop Points	Results
1	A				
	B				
	C				
	D				

c. Testing the Whole System

Table 3 below shows the mapping between banana quality classifications and the corresponding labels used in the system based on Philippine banana quality criteria for banana ketchup manufacturing. Bananas categorized as "Extra Class," "Class 1," or "Class 2" are labeled as "Good," indicating they meet the quality standards for processing. Bananas that fall outside these acceptable classes ("Excluded in the Class Standard") are labeled as "Bad," signifying they are unsuitable for processing.

Table 3. Banana Quality Classification

Banana Class	Classification Label
Extra Class, Class 1, Class 2	Good
Excluded in the Class Standard	Bad

III. RESULTS AND DISCUSSION



Figure 5. Prototype Setup

A. Method of Evaluation

To assess the detection capabilities of the YOLOv11 model, several classification metrics were employed, including accuracy, recall, precision, F1-score, and mean average precision (mAP). Label Studio provided accuracy, recall, and precision metrics following dataset training. To further evaluate the model's performance, the F1-score and mean average precision (mAP) were manually computed using the appropriate formulas. These metrics provided a comprehensive understanding of the model's effectiveness in detecting and classifying objects with precision and reliability.

Table 4. Performance Metrics of YOLOv11m

Performance Metrics						
Epochs	Class	Instances	Precision (%)	Recall (%)	mAP (%)	F1 Score (%)
100	All	157	99.8	99.4	99.4	99.1
	Bad	71	99.7	100	99.5	99.8
	Good	86	99.9	98.8	99.3	99.3

B. Confusion Matrix

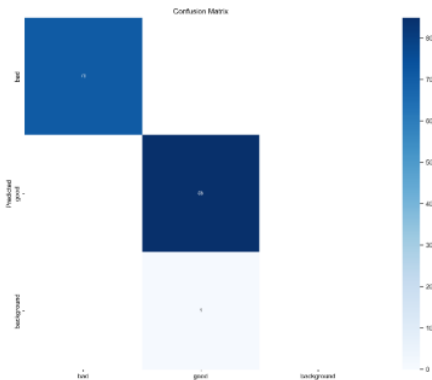


Figure 6. Confusion Matrix of YOLOv11m

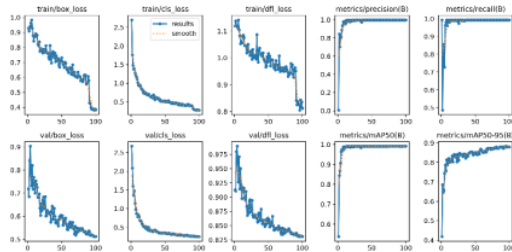


Figure 7. Results of YOLOv11m Training

To evaluate the robotic arm's banana classification accuracy, a confusion matrix of the trained YOLOv11m model was employed. This matrix effectively summarized the model's performance by quantifying four possible outcomes: True Positives (TP), where bananas were correctly

classified as bananas; True Negatives (TN), where non-banana objects were correctly classified as non-bananas; False Positives (FP), where non-banana objects were incorrectly classified as bananas (false alarms); and False Negatives (FN), where bananas were incorrectly classified as non-bananas (missed detections). This analysis was essential in determining the system's ability to accurately categorize bananas based on their quality, such as ripe, unripe, or defective. The insights derived from the confusion matrix provided a detailed understanding of both correct and incorrect classifications, helping to identify areas for improvement within the sorting algorithm.

C. Testing Validation

To validate the sorting system, a series of tests were conducted using randomly selected batches of bananas, categorized by quality attributes. The results of these experiments were analyzed using established performance equations, enabling a comprehensive evaluation of the robotic arm's performance compared to manual sorting methods. The systematic approach ensured a thorough examination of all aspects of the sorting process, leading to reliable conclusions regarding the system's effectiveness.

Table 5. Pick and Drop Points Testing Results

Trial	Classification		Pick		Drop	
	Actual	Recognized	Pick Points	Results	Drop Points	Results
1	Good	Good	A	A	G Container	Same
2	Bad	Bad	A	A	B Container	Same
Overall System Training Result						100%

The results of the system evaluation demonstrate the high accuracy of the robotic banana sorter in both classification and handling. In both trials, the system correctly recognized the actual classification of the bananas, successfully picking them from designated points and dropping them into appropriate containers. The "Good" banana was accurately placed in the good container, while the "Bad" banana was correctly sorted into the Bad container. With an overall system training result of 100%, these findings confirm the system's reliability and precision in classifying and sorting bananas, ensuring effective automation of the sorting process.

D. Comparison of YOLOv11 Models

Table 6. Performance Evaluation of YOLOv11

Performance Metrics							
Model	Epochs	Class	Instances	Precision (%)	Recall (%)	mAP (%)	F1 Score (%)
YOLO11n	100	All	157	99.9	99.4	99.1	99.6
		Bad	71	99.9	100	99.5	99.9
		Good	86	99.9	98.7	98.7	99.3
YOLO11m		All	157	99.8	99.4	99.4	99.1
		Bad	71	99.7	100	99.5	99.8
		Good	86	99.9	98.8	99.3	99.3
YOLO1		All	157	99.3	99.4	99.3	99.3

1x	Bad	71	99.8	100	99.5	99.9
	Good	86	98.7	98.8	99.2	99.2

The choice of the YOLOv11 model variant was carefully considered to balance speed, accuracy, and hardware compatibility for the task at hand. The YOLOv11n variant was not utilized due to its high-speed performance being offset by low accuracy, making it unsuitable for precise object detection. Similarly, the YOLOv11s variant, while offering high accuracy, required higher GPU specifications to achieve optimal performance. Although the current GPU, an NVIDIA® GeForce RTX™ 4070 with 8GB GDDR6, is capable, its memory capacity posed limitations for efficient handling of YOLOv11s. Ultimately, the YOLOv11m variant was selected as it provided an optimal balance between speed and accuracy while aligning with the available hardware capabilities, meeting the requirements of both efficiency and precision in the deployment environment.

E. Testing the Whole System

Table 7. AI Classification Testing Results

Trial	Input Description	Pickup Result	Drop Result	Accuracy
1	Extra Class, Class 1, Class 2	A	A	100%
2	Excluded in Class Standard	A	A	100%
3	Extra Class, Class 1, Class 2	B	B	100%
4	Excluded in Class Standard	B	B	100%
5	Extra Class, Class 1, Class 2	C	C	100%
6	Excluded in Class Standard	C	C	100%
7	Extra Class, Class 1, Class 2	D	D	100%
8	Excluded in Class Standard	D	D	100%
Percentage of Correct Classification				100%

The results of the trials indicate that the robotic arm consistently achieved perfect accuracy in sorting bananas. In all eight trials, the system correctly picked up and dropped bananas into their respective categories—Extra Class, Class 1, Class 2, and Excluded in Class Standard—resulting in an accuracy of 100% for each trial. This demonstrates the system's reliability and precision in categorizing bananas based on their quality attributes, confirming its effectiveness in the sorting process. The overall percentage of correct classifications was 100%, highlighting the robotic arm's high performance in comparison to manual sorting methods.

IV. CONCLUSION

Conclusion

The study successfully developed a robotic system for automated quality control in banana ketchup manufacturing, enhancing efficiency, reducing manual labor, and minimizing human error in banana sorting. A key innovation was the deployment of a robust computer vision model that

accurately identified and classified "Saba" bananas into four freshness categories (ripe, unripe, overripe, and defective) based on diverse image cues such as edges, texture, shape, and color. Integrated with a robotic arm, this system enabled precise pick-and-place operations, demonstrating the feasibility of automation in banana handling.

Performance evaluation of YOLOv11 model variants (YOLOv11n, YOLOv11m, and YOLOv11x) revealed trade-offs in speed, accuracy, and hardware efficiency. YOLOv11n, though the fastest, exhibited lower accuracy in classifying "Good" bananas. YOLOv11x showed marginally improved recall but had higher computational demands. YOLOv11m emerged as the optimal choice, balancing speed and accuracy with a precision of 99.8%, recall of 99.4%, and mAP of 99.4%. It excelled in both "Good" and "Bad" classifications while maintaining compatibility with the NVIDIA® GeForce RTX™ 4070 GPU, making it the ideal model for real-time deployment.

This research marks a significant advancement in automated banana quality control, providing a scalable and reliable solution for banana ketchup manufacturing while setting a foundation for future innovations in agricultural and food processing technologies.

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